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# Step 1: Import required libraries
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Step 2: Upload an image
uploaded = files.upload()

# Step 3: Read the uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)

# Step 4: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 5: Apply Sobel Edge Detection along Y-axis
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Step 6: Convert result to absolute scale
sobel_y = cv2.convertScaleAbs(sobel_y)

# Step 7: Display original and Sobel Y result
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Edge Detection - Y Axis")
plt.axis("off")

plt.show()
```

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Original Image

Sobel Edge Detection - Y Axis